

TASK 9A

9A-Fetch data from a REST APL

In this program, Flutter sends a GET request to a REST API, receives JSON data, converts it using `jsonDecode()`, and displays the user details using `ListView.builder`.

“We requested user data from a REST API using a GET request. The server returned the data in JSON format. We decoded the JSON using `jsonDecode()`, stored it in a list, and displayed the names and emails using `ListView.builder`. Therefore, the data shown on our Flutter screen comes from the REST API.”

REST = REpresentational State Transfer

API = Application Programming Interface

It is an architectural style that uses HTTP protocol to communicate between Client and Server. It is stateless.

Key Features

Uses JSON as data format (earlier XML)

Works on HTTP/HTTPS

Gives Status Codes: 200 OK, 201 Created, 400 Bad Request, 404 Not Found, 500 Server Error

Lightweight and Fast - used in Mobile Apps, Web Apps

JSON = JavaScript Object Notation

It's a lightweight format for storing and transporting data.

It's how a Client and Server talk to each other in a REST API.

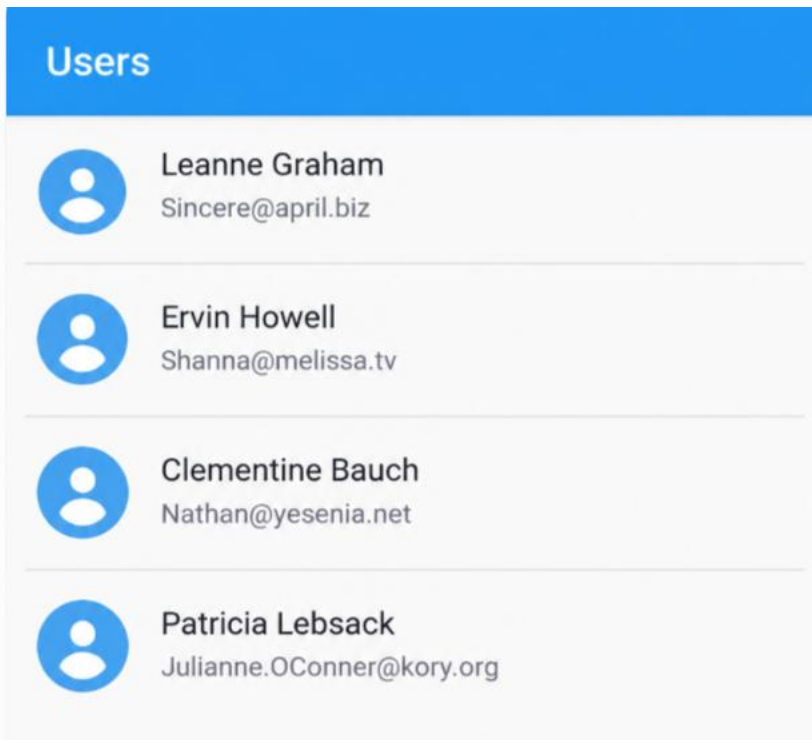
Syntax

It is written as Key : Value pairs inside { }

Data is in name : value pair

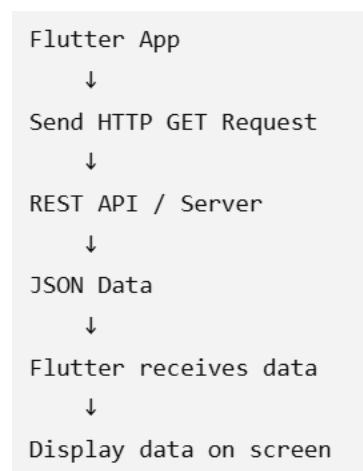
Keys must be in double quotes " "

Values can be String, Number, Boolean, Array, Object, null



A REST API is a way for our Flutter application to communicate with a server.

We send a request to the server, the server sends data back, usually in **JSON format**, and Flutter displays that data on the screen.



Fetch User Data:

use a simple public API: It provides sample user information.

<https://jsonplaceholder.typicode.com/users>

Send GET request

```
final response = await http.get(
  Uri.parse('https://jsonplaceholder.typicode.com/users'),
);
```

The API returns data like:

```
[
  {
    "name": "Leanne Graham",
    "email": "Sincere@april.biz"
  }
]
```

Display the data

```
ListView.builder(
```

We use `ListView.builder` to display multiple users.

```
title: Text(users[index]['name']),
subtitle: Text(users[index]['email']),
```

This displays the user's **name and email**.

In the Flutter project terminal:

RUN:

```
flutter pub add http
```

PROGRAM:9A

```
import 'dart:convert';
```

```
import 'package:flutter/material.dart';
```

```
import 'package:http/http.dart' as http; // his package helps Flutter send HTTP
requests to the API.
```

```
void main() {
```

```
  runApp(const MyApp());
```

```
}
```

```
class MyApp extends StatelessWidget {  
  const MyApp({super.key});  
  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: const UserPage(),  
      debugShowCheckedModeBanner: false,  
    );  
  }  
}
```

```
class UserPage extends StatefulWidget {  
  const UserPage({super.key});  
  
  @override  
  State<UserPage> createState() => _UserPageState();  
}
```

```
class _UserPageState extends State<UserPage> {  
  List users = []; // We create an empty list to store the data received from the API.
```

```
Future<void> fetchUsers() async {
```

```
  final response = await http.get( // asks the server for data.
```

```
    Uri.parse('https://jsonplaceholder.typicode.com/users'),
  );
```

```
  if (response.statusCode == 200) { //Check response- Status code 200 means
the request was successful.
```

```
    setState(() {
```

```
      users = jsonDecode(response.body); // The server sends JSON. jsonDecode ()
converts that JSON into data that Dart can use.
```

```
    });
```

```
  }
```

```
}
```

```
@override
```

```
void initState() {
```

```
  super.initState();
```

```
  fetchUsers();
```

```
}
```

```
@override
```

```
Widget build(BuildContext context) {
```

```
  return Scaffold(
```

```
    appBar: AppBar(
```

```
      title: const Text('Users'),
```

```
),  
body: ListView.builder( // to display multiple users.  
  itemCount: users.length,  
  itemBuilder: (context, index) {  
    return ListTile(  
      title: Text(users[index]['name']),  
      subtitle: Text(users[index]['email']),  
    );  
  },  
),  
);  
}  
}
```